

Predicting Eruptive Events at Volcanoes from Earthquake Data

UE24CS352A Machine Learning Mini-Project - Section E
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Problem and objective

Volcanic unrest often changes the local earthquake pattern. This project asks whether an earthquake catalog can identify the eruptive state of Kilauea's Pu'u 'O'o system. We frame the task as contemporaneous binary classification: for each catalog earthquake, predict whether it occurred during a documented eruptive interval. This is a retrospective study and should not be read as an operational warning system.

Dataset and target construction

The project uses 6,485 ANSS/WOVOdat earthquake records from the Pu'u 'O'o area, spanning January 1983 to December 1986. Each record contains timestamp, latitude, longitude, depth, magnitude, and distance from the vent. We read the detailed Pu'u 'O'o eruption chronology from the accompanying Hawaii Center for Volcanology workbook. A label of 1 means the earthquake timestamp falls inside an eruption interval; otherwise the label is 0. The resulting positive rate is 68.9%.

Features and methodology

The model uses latitude, longitude, depth, magnitude, distance, and earthquake-history features. For each event, the pipeline counts earlier earthquakes and records the maximum earlier magnitude in the previous 1, 7, and 30 days. The current earthquake is excluded from these rolling features. We use a chronological split: 70% training, 15% development, and 15% held-out testing. This avoids allowing future records to inform past examples. We compare a majority baseline, balanced Logistic Regression with standardization, and a balanced Random Forest with 300 trees and maximum depth 15.

Results

Model	Accuracy	Precision	Recall	F1	Kappa	ROC-AUC
Majority Baseline	0.550	0.550	1.000	0.710	0.000	0.500
Logistic Regression	0.475	1.000	0.045	0.086	0.041	0.961
Random Forest	0.636	0.882	0.391	0.541	0.308	0.822

The held-out ROC-AUC is useful because the test period has a different event-rate distribution from training. Logistic Regression reaches ROC-AUC 0.961, while Random Forest reaches 0.822; both outperform the 0.5 baseline when ranking eruptive cases. We select thresholds on the development period rather than tuning on the test set. The Random Forest threshold is 0.75, giving test accuracy 0.636, F1 0.541, and Cohen's kappa 0.308.

Interpretation and limitations

Earthquake catalog features contain information associated with Kilauea's eruptive state, especially in the ranking measured by ROC-AUC. Development-set threshold selection improves the Random Forest's balance between false alarms and missed eruptive events, but the result remains sensitive to the time period and label definition. A stronger follow-up should use event-based validation and add waveform/tremor, GPS deformation, gas, and station-quality features. The present model predicts contemporaneous status rather than the time until a future eruption, so it supports a useful educational demonstration but not public safety decisions.

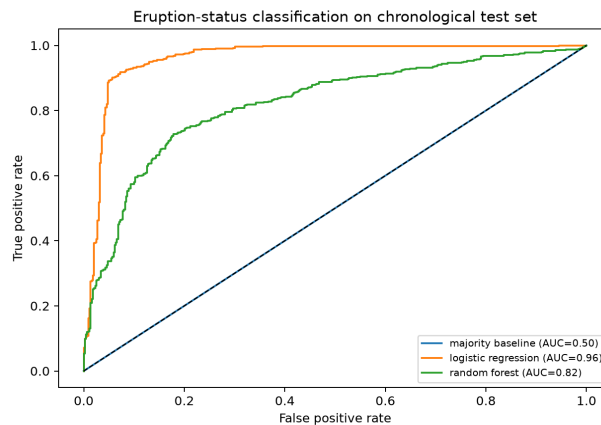


Figure: ROC curves on the held-out chronological test period.

Sources: supplied CS229 report and public reference repository [bmullet/PEEVED](https://github.com/bmullet/PEEVED); ANSS/WOVOdat earthquake catalog; Hawaii Center for Volcanology Kilauea eruption chronology.